

Network Threats and Mitigation & Physical Security and Risk

Task 1 :- Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

ANS :-

Network Threat	Real-World Example
Phishing	A fake SMS asking you to click a link to update your Paytm KYC to stop account block.
DDoS	Attackers overload Instagram servers with fake traffic, making the app crash for everyone.
Man-in-the-Middle	A hacker on free public Wi-Fi intercepts your WhatsApp messages before they reach the server.
Malware	Downloading a fake WhatsApp Gold APK file that sneaks a virus onto your phone.
Password Attack	A hacker uses automated scripts to guess your Instagram password until they get it right.

Task 2 :- Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.

ANS :-

```
SECRET_PIN = "5829"

# Try all combinations from 0000 to 9999
for i in range(10000):
    guess = f"{i:04d}" # Formats integer as 4 digits (e.g., 7 -> "0007")

    if guess == SECRET_PIN:
        print(f"PIN Cracked: {guess}")
        break
```

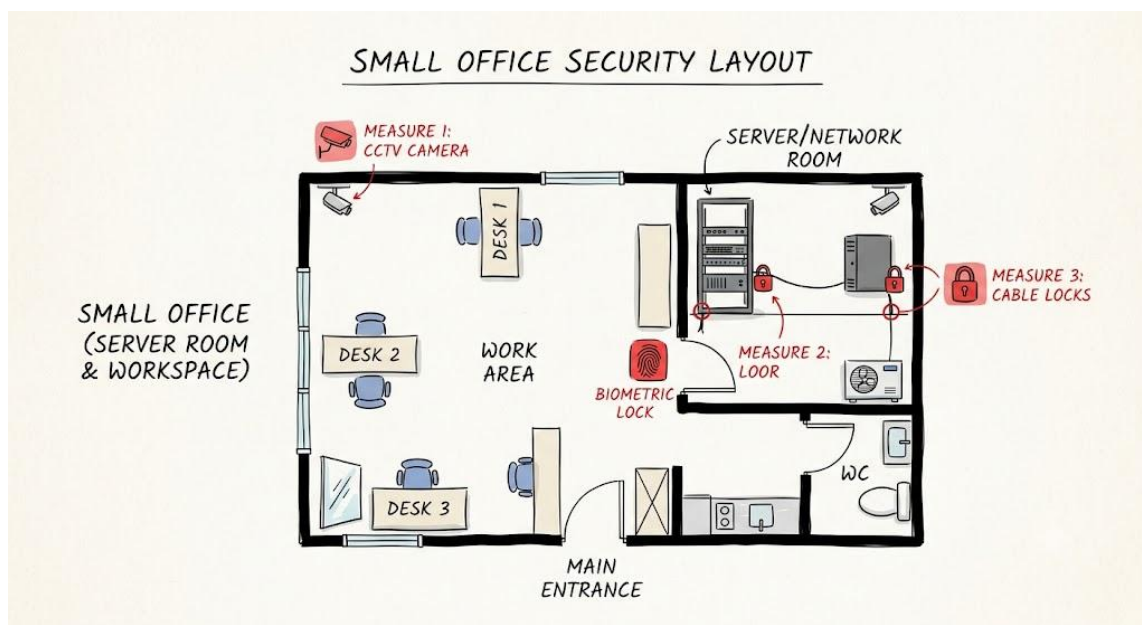
Task 3 :- List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

ANS :-

Physical Security Risk	Mitigation Step
Unauthorized Access	Use biometric scanners or smart keycards at the door.
Fire Damage	Install automated gas-based fire suppression systems.
Overheating	Maintain redundant precision air conditioning units (HVAC).
Power Loss	Use backup UPS systems and industrial diesel generators.

Task 4 :- Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

ANS :-



Task 5 :- Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

ANS :-

College Computer Lab Checklist :-

1.  Use strong passwords and MFA.
2.  Keep antivirus and firewalls on.
3.  Update lab computers regularly.
4.  Secure Wi-Fi and network access.
5.  Give access only to authorized students/staff.
6.  Lock server rooms and network equipment.
7.  Use cameras and check lab entry/exit.